Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

Gajavada Sanjeevkumar

Problem Statement

Question

Draw a **cycloid** generated by a circle of diameter $D = 50$ mm rolling along a straight base line for **one complete revolution** without slipping.

Construct a **tangent** and a **normal** to the cycloid at any point P located on the curve.

Step 1: Draw Generating Circle & Directing Line (Base)

C_0 $\longrightarrow$ Initial Center Point
AA' $\longrightarrow$ Directing Line
Axis $\longrightarrow$ Center Line

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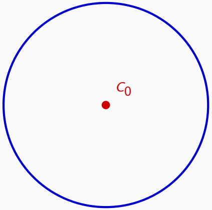

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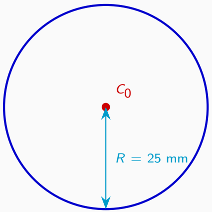

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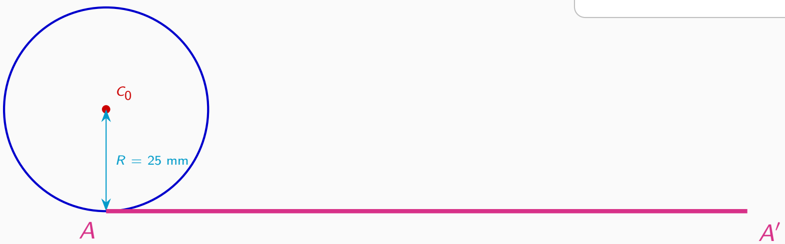

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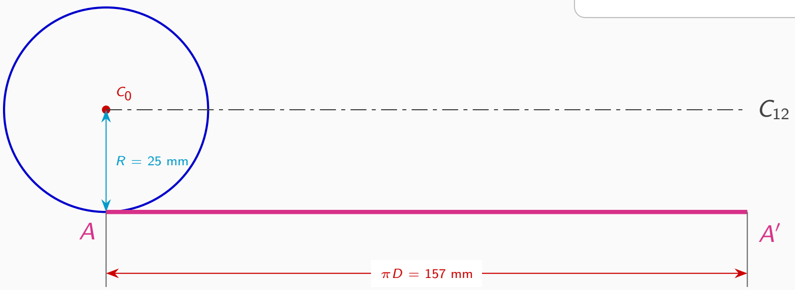

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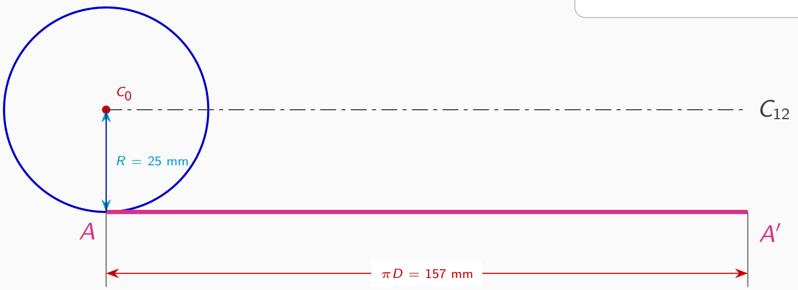
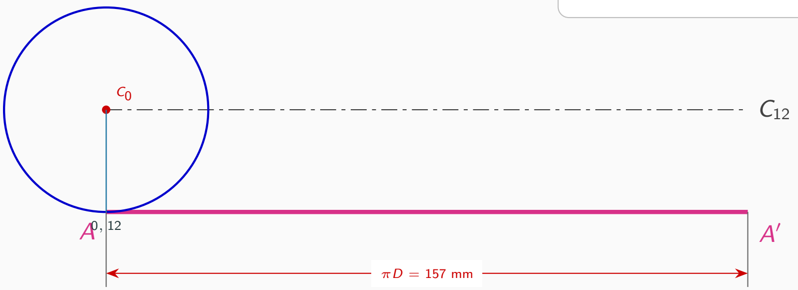


Step 2: Divide Generating Circle into 12 Equal Parts

AA'	$\longrightarrow$ Directing Line
Axis Line	$\longrightarrow$ Center Line
$0 \ldots 12$	$\longrightarrow$ Circle Divisions



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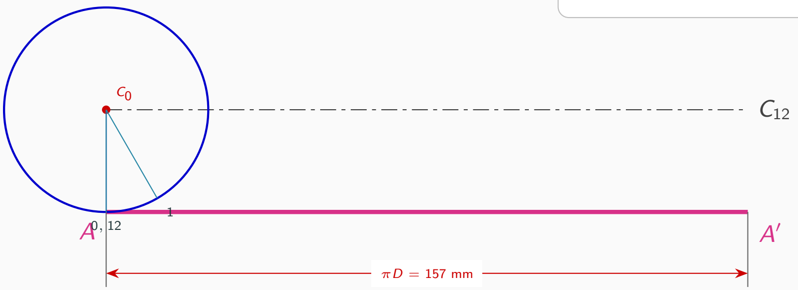

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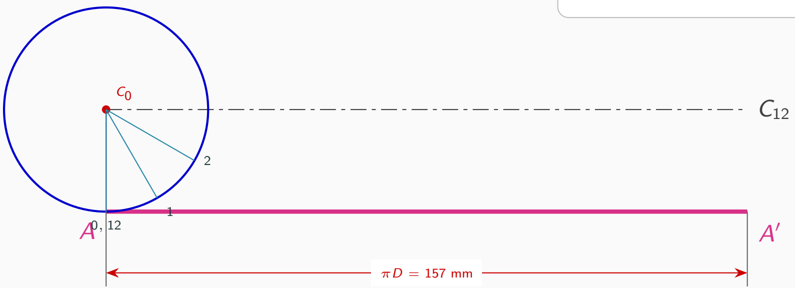

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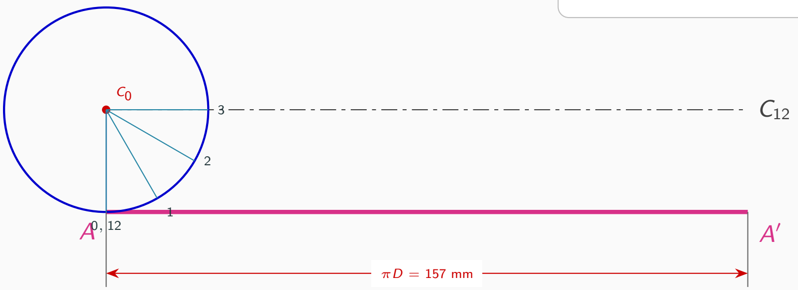

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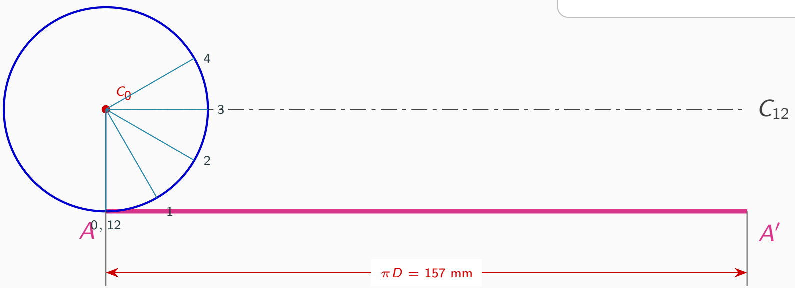

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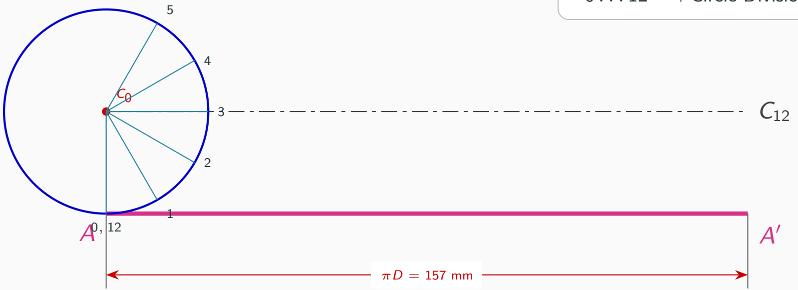

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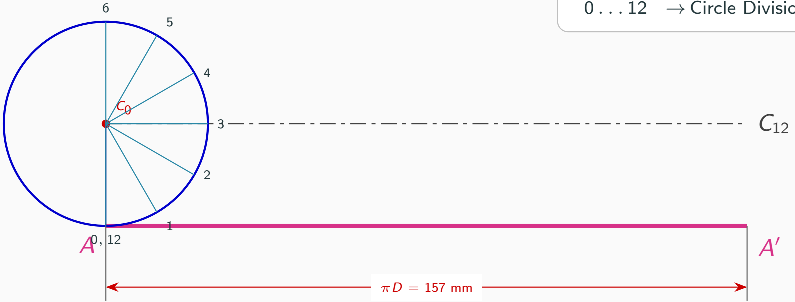

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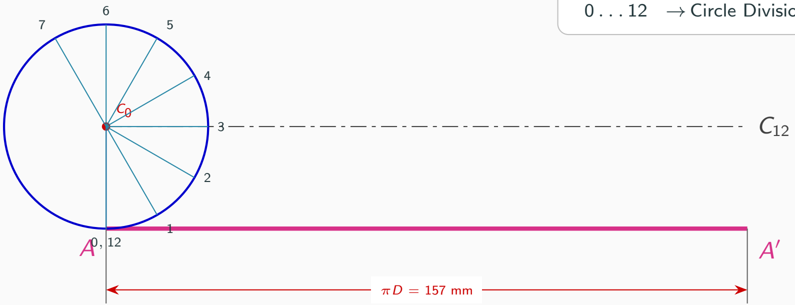

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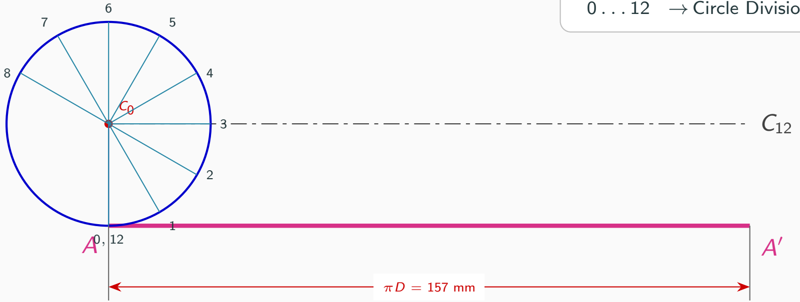

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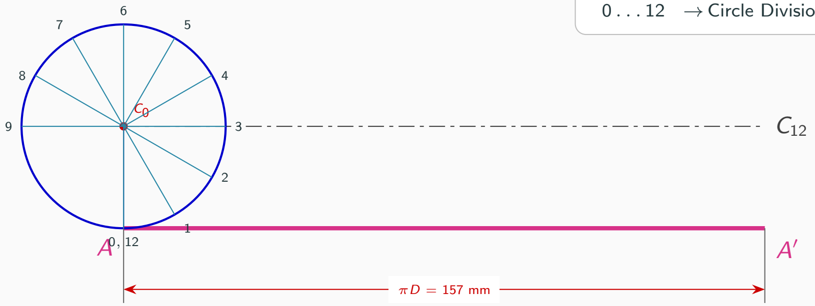

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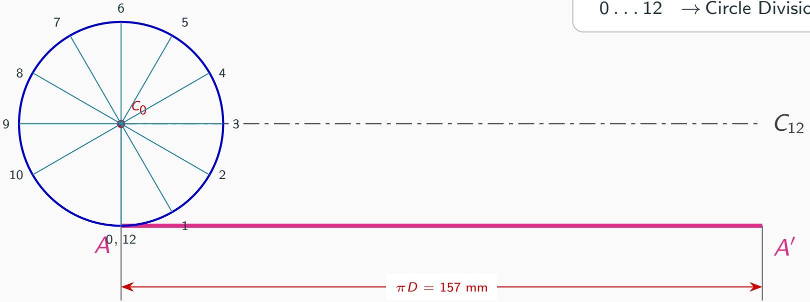

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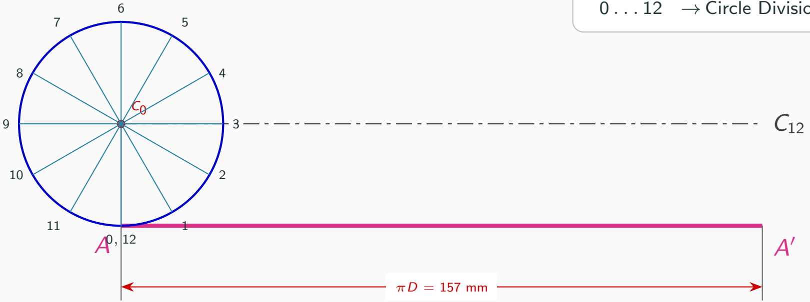
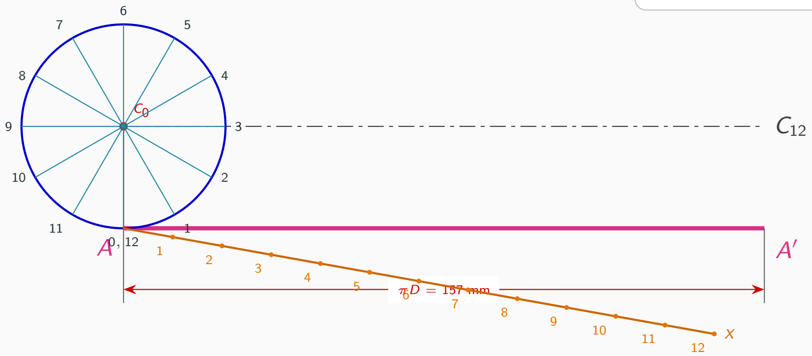


AA'	$\rightarrow$ Directing Line
Axis Line	$\rightarrow$ Center Line
$0 \ldots 12$	$\rightarrow$ Circle Divisions

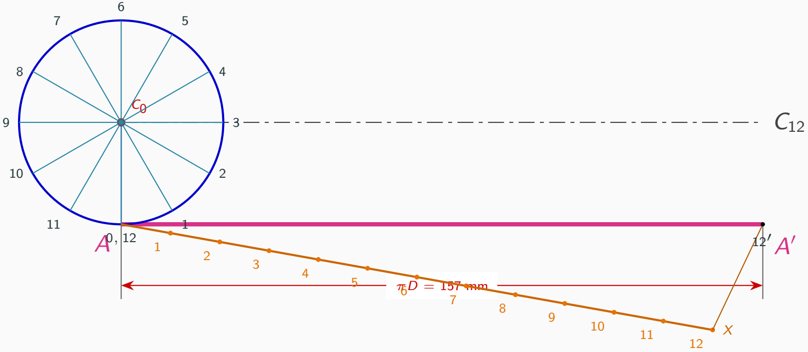
Step 3: Divide Directing Line - Setup Inclined Line AX

$AA' \rightarrow$ Directing Line
$AX \rightarrow$ Inclined Line



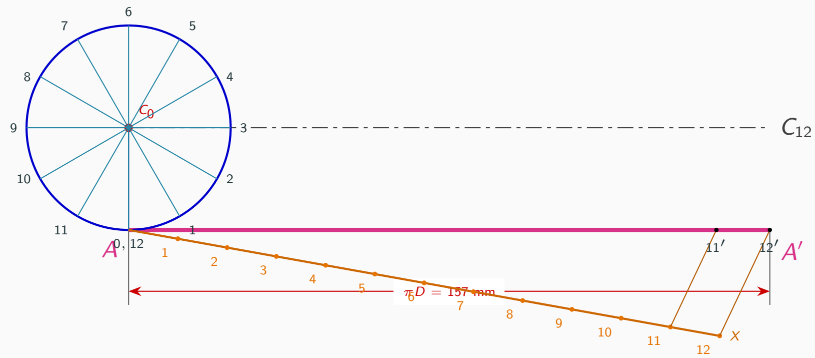
Step 3: Divide Directing Line - Transfer Point 12'

$12' \rightarrow$ Transfer Point 12



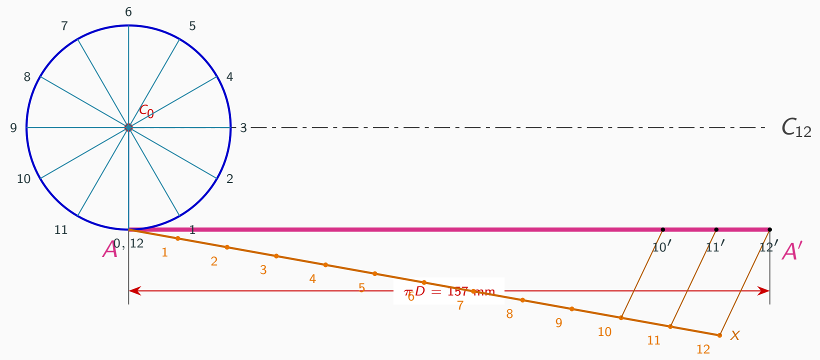
Step 3: Divide Directing Line - Transfer Point 11'

$11' \rightarrow$ Transfer Point 11



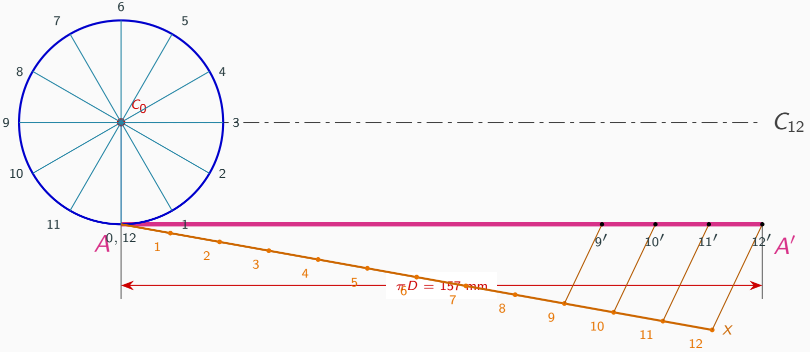
Step 3: Divide Directing Line - Transfer Point 10’

$10' \rightarrow$ Transfer Point 10



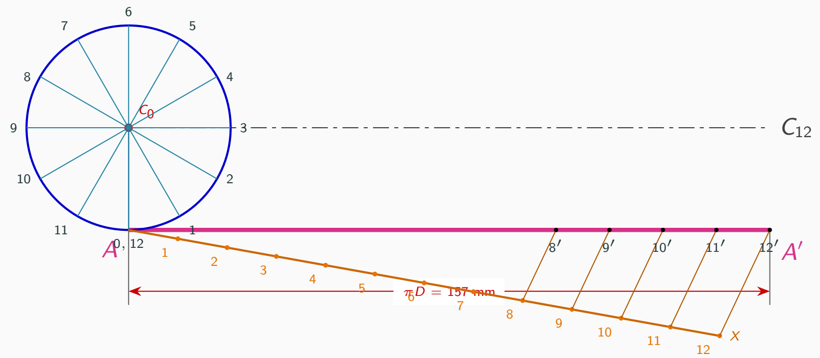
Step 3: Divide Directing Line - Transfer Point 9'

$9' \rightarrow$ Transfer Point 9



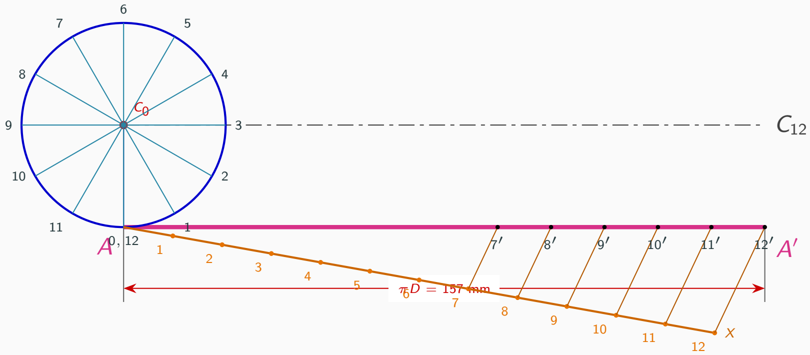
Step 3: Divide Directing Line - Transfer Point 8'

$8' \rightarrow$ Transfer Point 8



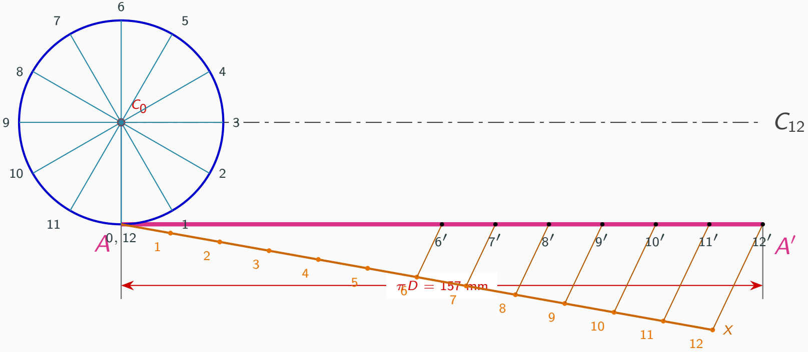
Step 3: Divide Directing Line - Transfer Point 7'

$7' \rightarrow$ Transfer Point 7



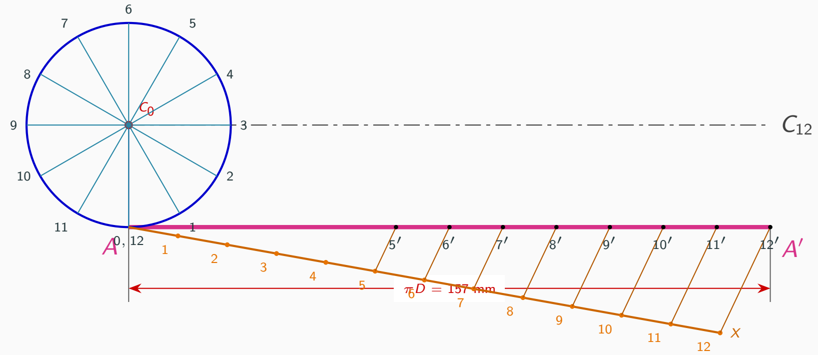
Step 3: Divide Directing Line - Transfer Point 6’

$6' \rightarrow$ Transfer Point 6



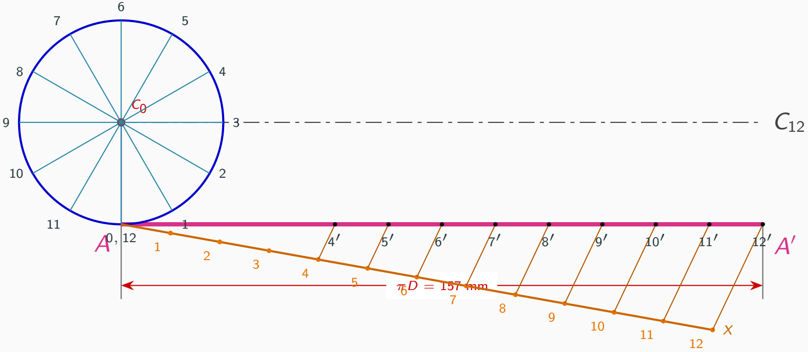
Step 3: Divide Directing Line - Transfer Point 5’

$5' \rightarrow$ Transfer Point 5



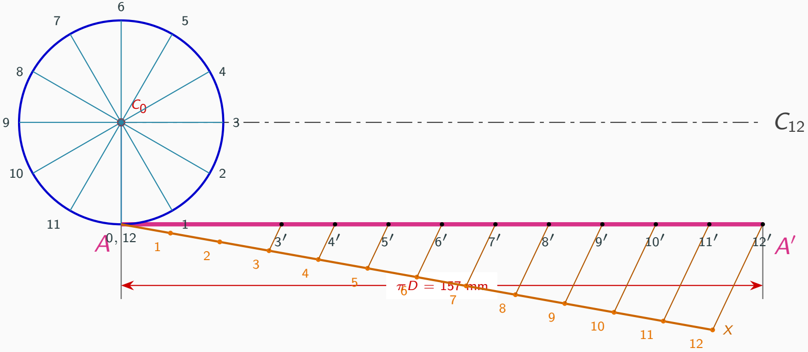
Step 3: Divide Directing Line - Transfer Point 4'

$4' \rightarrow$ Transfer Point 4



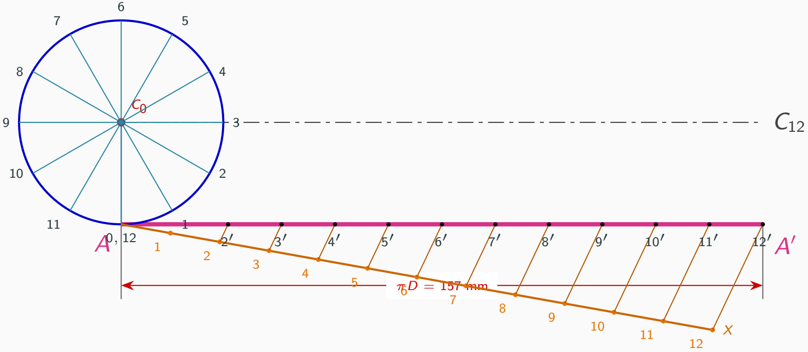
Step 3: Divide Directing Line - Transfer Point 3'

3′ → Transfer Point 3



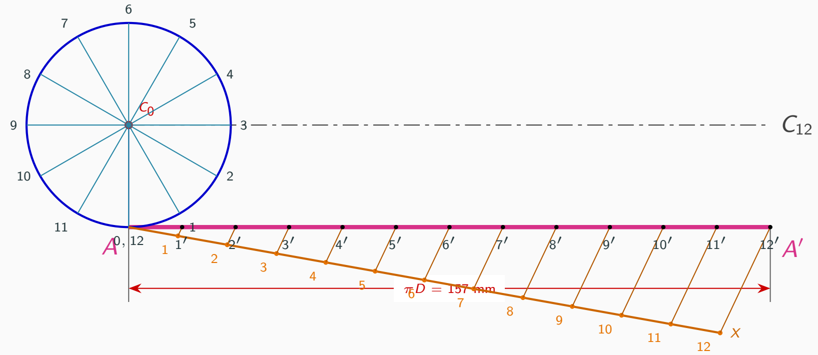
Step 3: Divide Directing Line - Transfer Point 2'

$2' \rightarrow$ Transfer Point 2



Step 3: Divide Directing Line - Transfer Point 1’

$1' \rightarrow$ Transfer Point 1



Step 4: Project Divisions to Locate Centers $C_1 \ldots C_{12}$ (Base)

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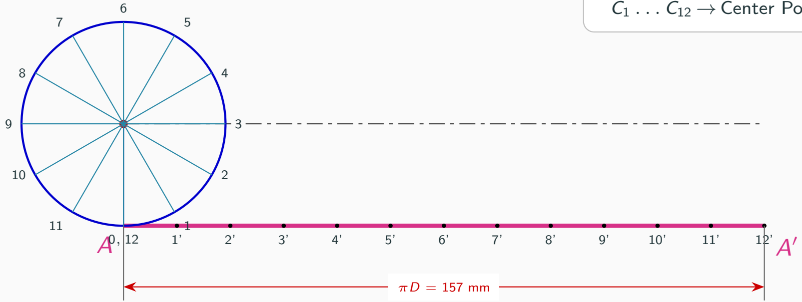

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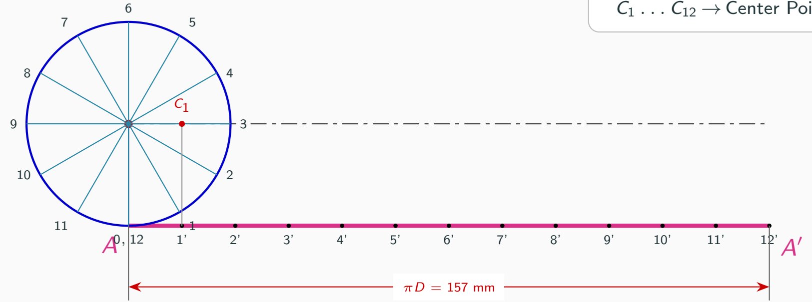

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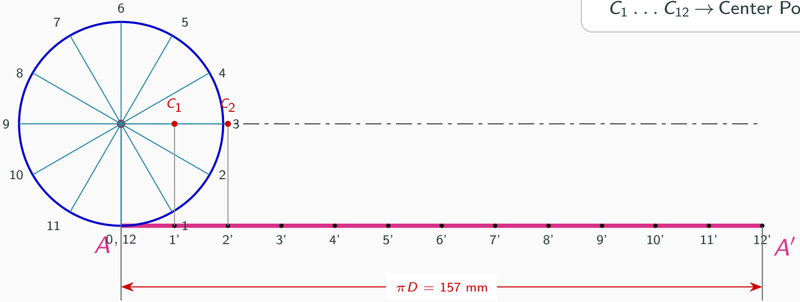

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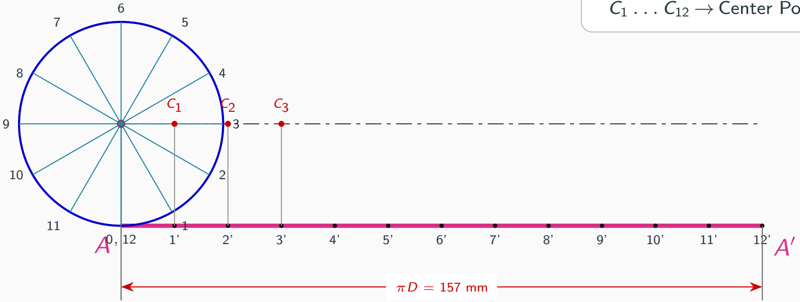

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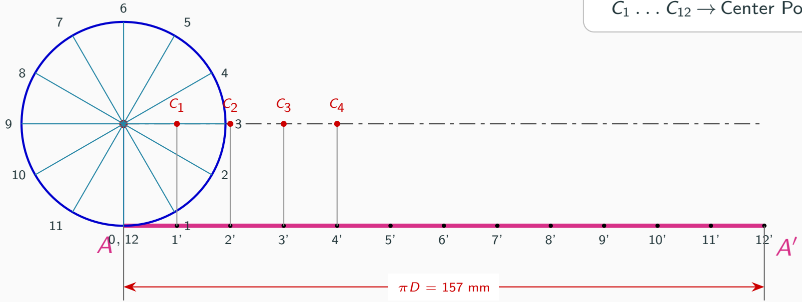

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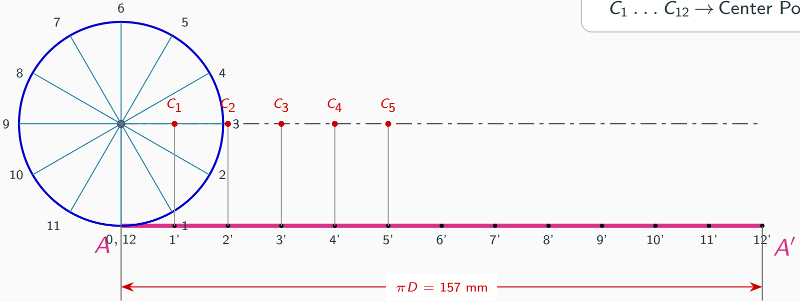

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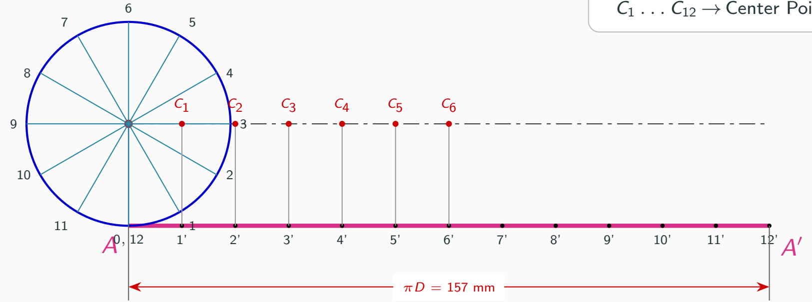

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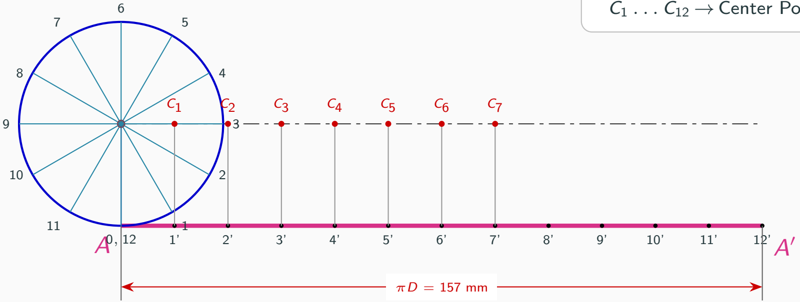

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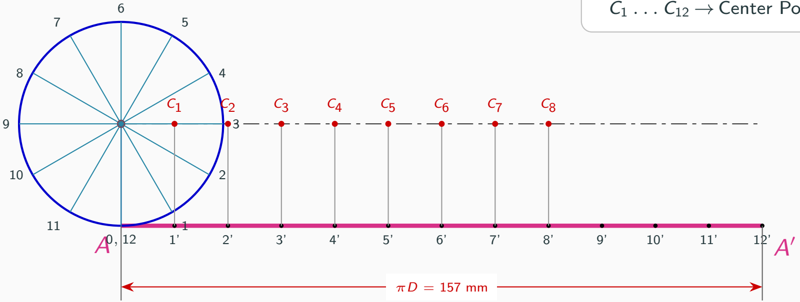

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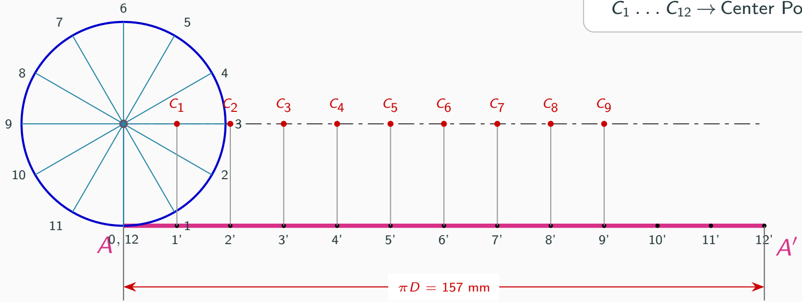

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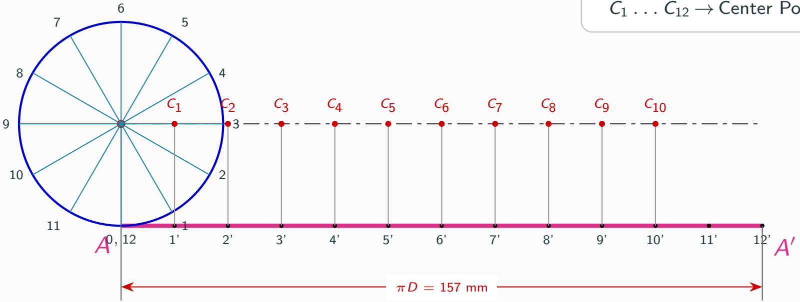

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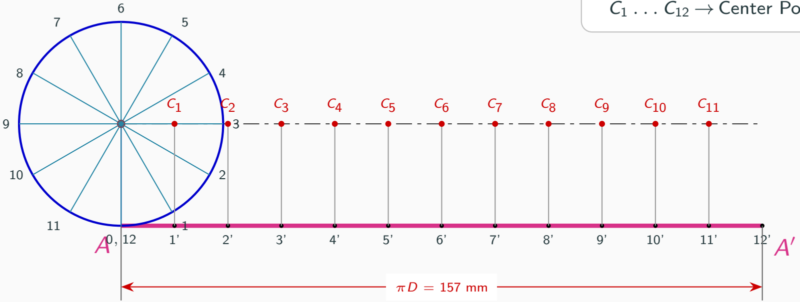

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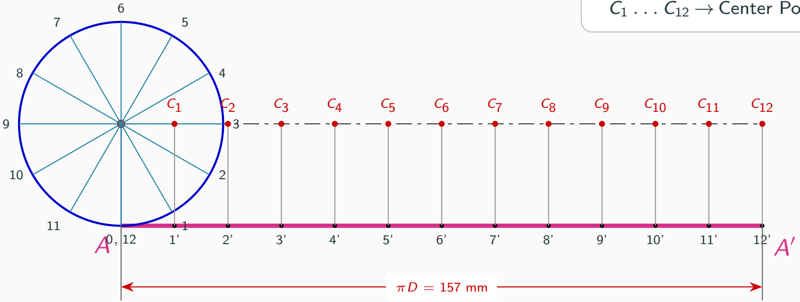

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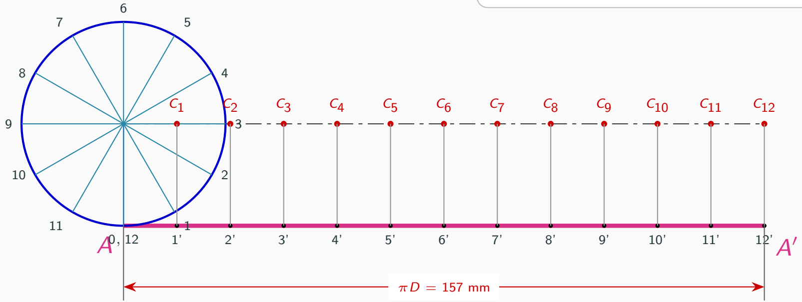
Step 5: Draw Horizontal Reference Lines (Base)

Blue Lines $\rightarrow$ Horizontal Reference Lines
$C_1 \ldots C_{12}$ $\rightarrow$ Retained Centers



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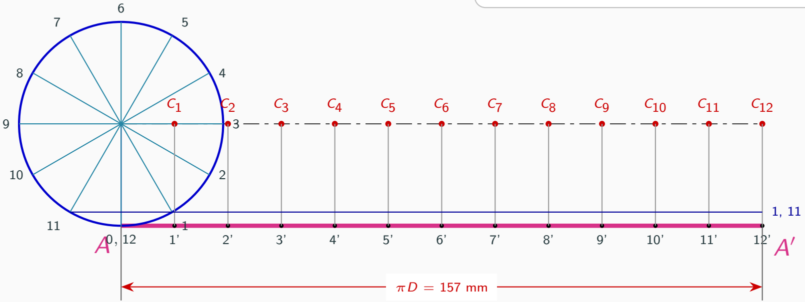

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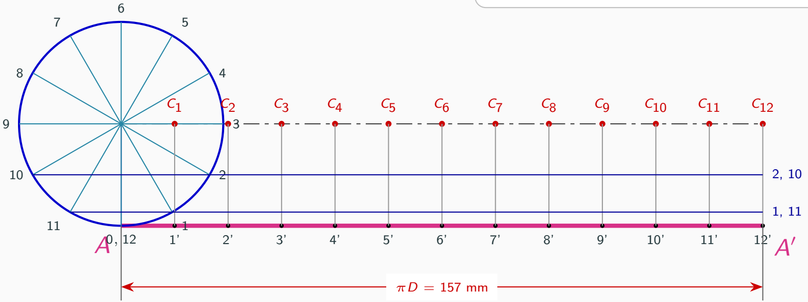

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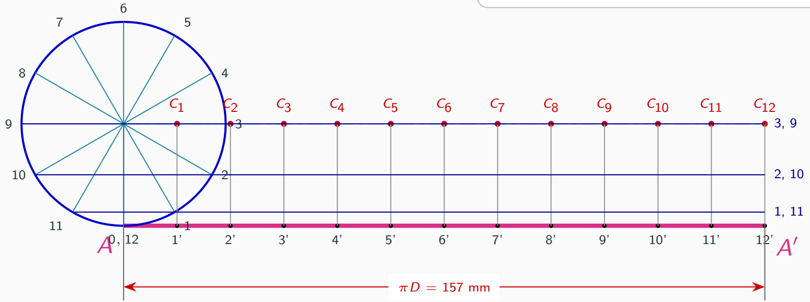

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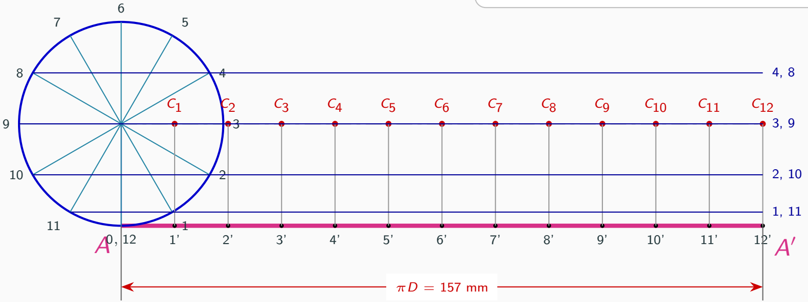

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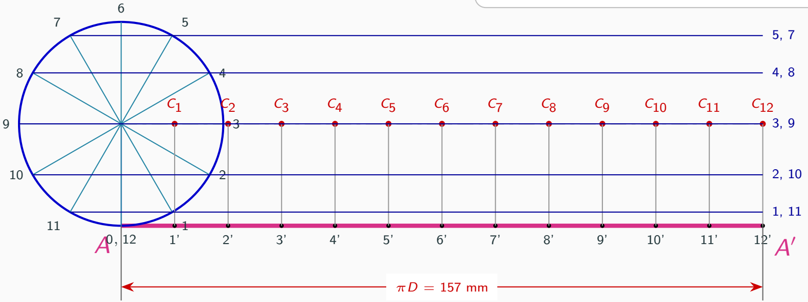

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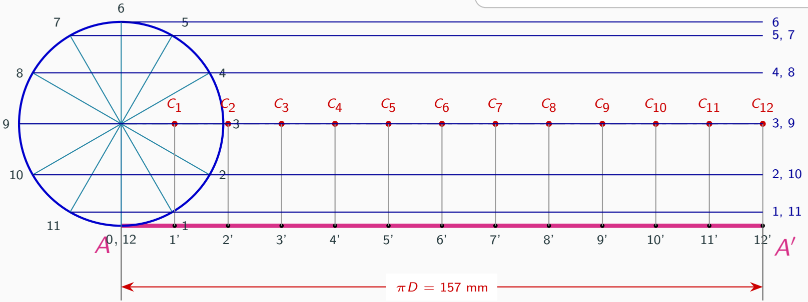
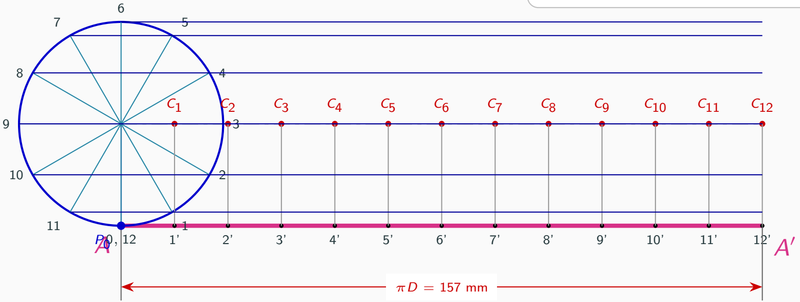
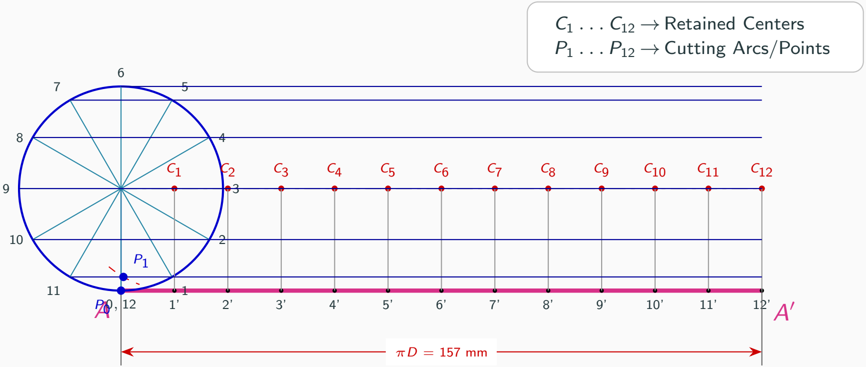


Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_0

$C_1 \ldots C_{12} \rightarrow$ Retained Centers
$P_1 \ldots P_{12} \rightarrow$ Cutting Arcs/Points

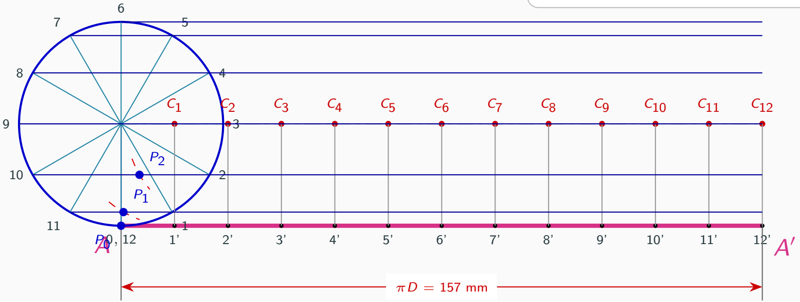


Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_1



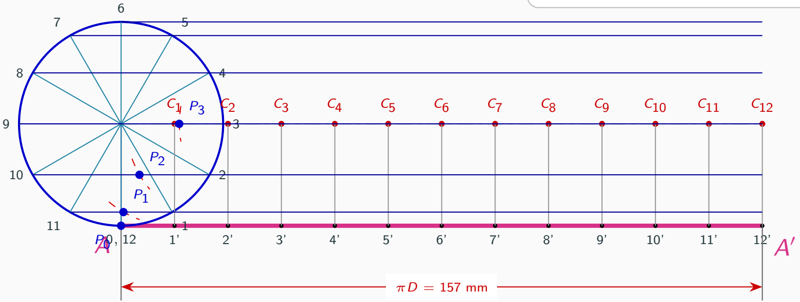
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_2

$C_1 \dots C_{12} \rightarrow$ Retained Centers
$P_1 \dots P_{12} \rightarrow$ Cutting Arcs/Points



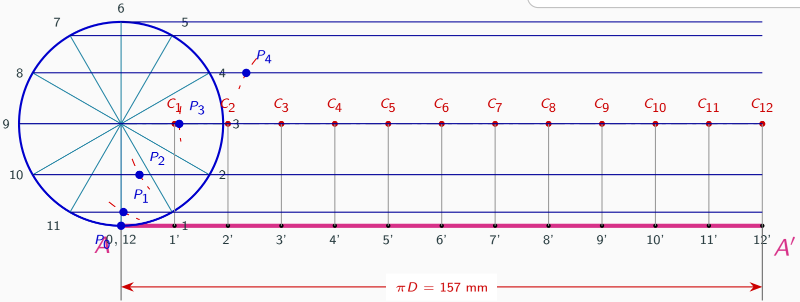
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_3

$C_1 \ldots C_{12} \rightarrow$ Retained Centers
$P_1 \ldots P_{12} \rightarrow$ Cutting Arcs/Points



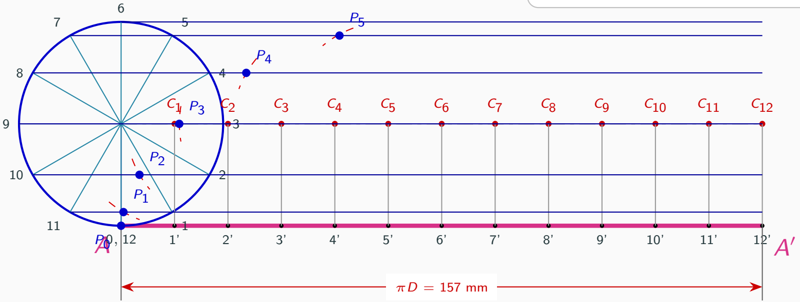
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_4

$C_1 \ldots C_{12} \rightarrow$ Retained Centers
$P_1 \ldots P_{12} \rightarrow$ Cutting Arcs/Points



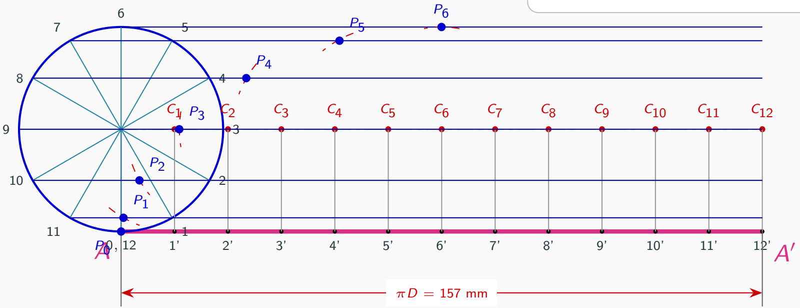
Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_5

$C_1 \ldots C_{12} \rightarrow$ Retained Centers
$P_1 \ldots P_{12} \rightarrow$ Cutting Arcs/Points



Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_6

$C_1 \ldots C_{12} \rightarrow$ Retained Centers
$P_1 \ldots P_{12} \rightarrow$ Cutting Arcs/Points



Step 6: Strike Cutting Arcs (Radius = 25 mm) to Locate P_7

$C_1 \ldots C_{12} \rightarrow$ Retained Centers
$P_1 \ldots P_{12} \rightarrow$ Cutting Arcs/Points